

Multivariable Calculus and its Applications

Functions of Several Variables

Definition:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

Examples:

- Loss function: $L(w_1, w_2, \dots, w_n)$
- Temperature map: $T(x, y)$
- $f(x, y) = x^2 + 3xy$

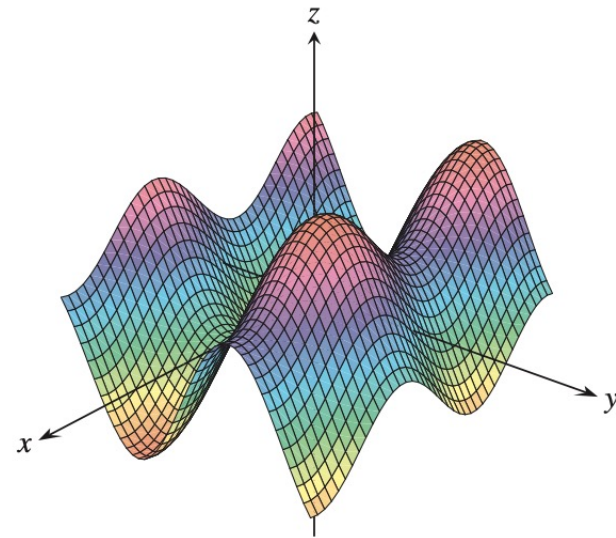
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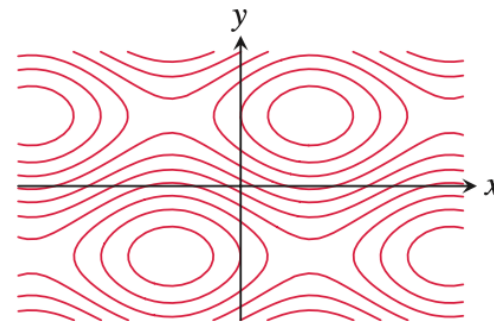
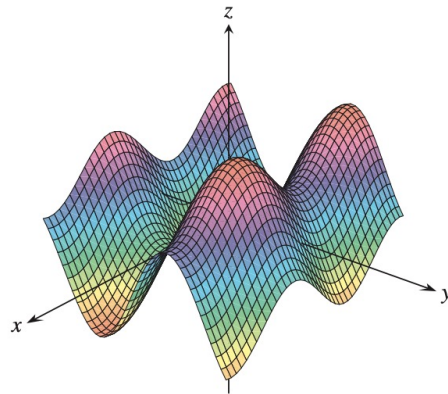


Functions of Several Variables

Level Curves:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

- The set of points in the plane where a function $f(x, y)$ has a constant value $f(x, y) = c$ is called a **level curve** of f .
- The set of all points $(x, y, f(x, y))$ in space, for (x, y) in the domain of f , is called the **graph** of f . The graph of f is also called the **surface $z = f(x, y)$** .

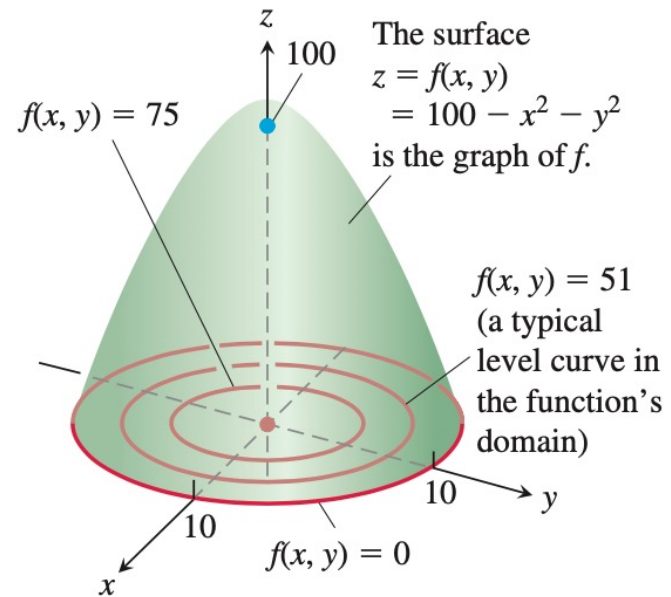


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Partial Derivatives

Derivative while holding other variables constant.

Example:

$$f(x, y) = x^2y + 5y$$
$$\frac{\partial f}{\partial x} = 2xy, \quad \frac{\partial f}{\partial y} = x^2 + 5$$

Interpretation:

- How sensitive output is to input changes
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Partial Derivatives

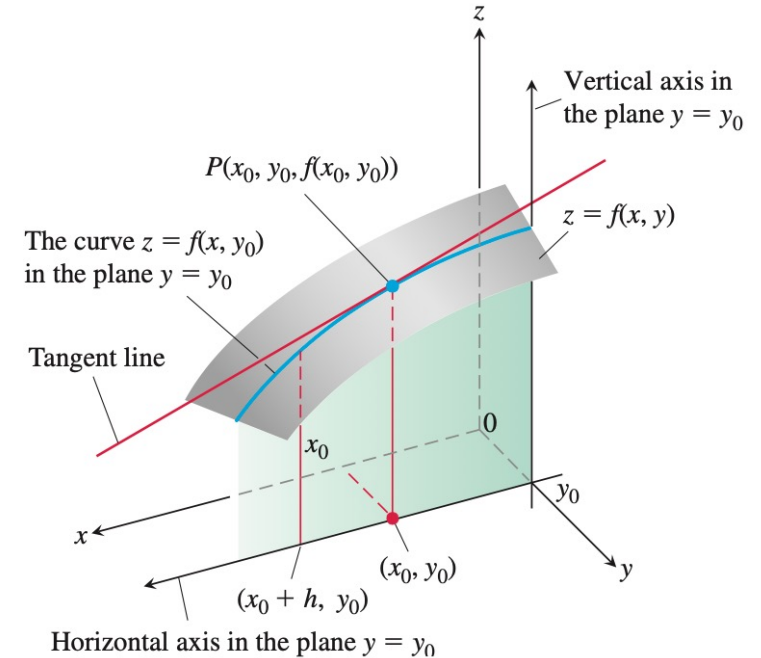
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Example:

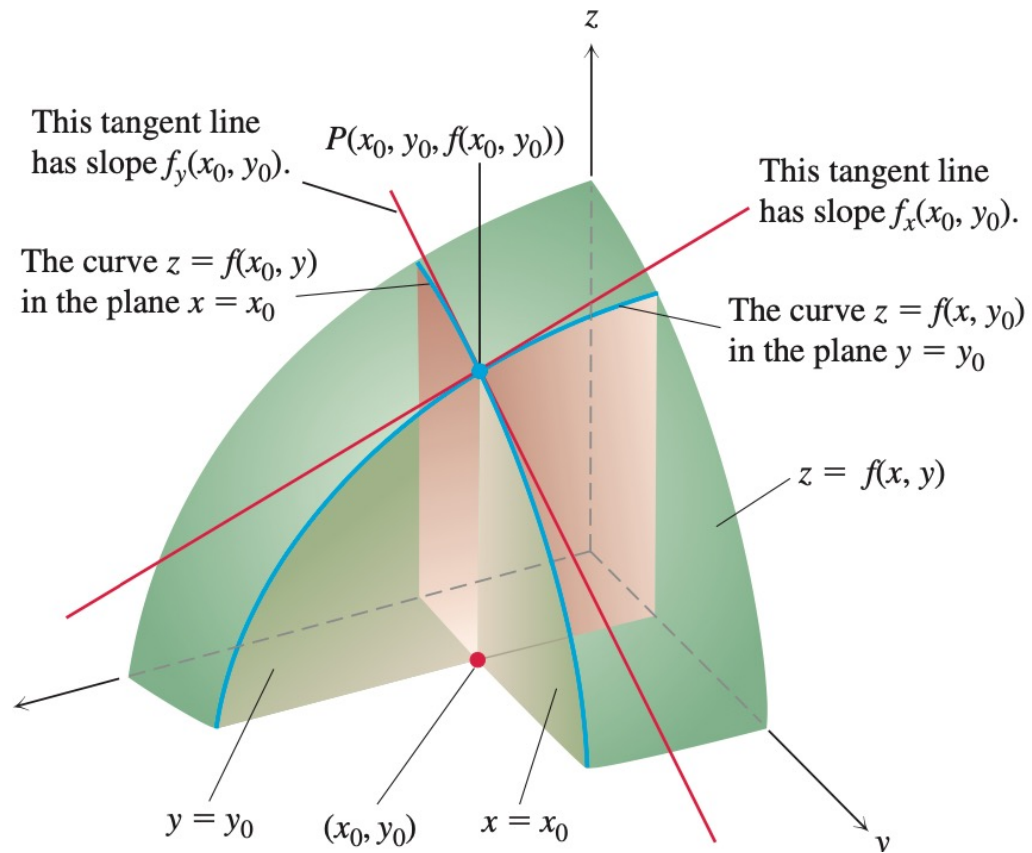
$$\begin{aligned} f(x, y) &= x^2y + 5y \\ \frac{\partial f}{\partial x} &= 2xy, & \frac{\partial f}{\partial y} &= x^2 + 5 \end{aligned}$$

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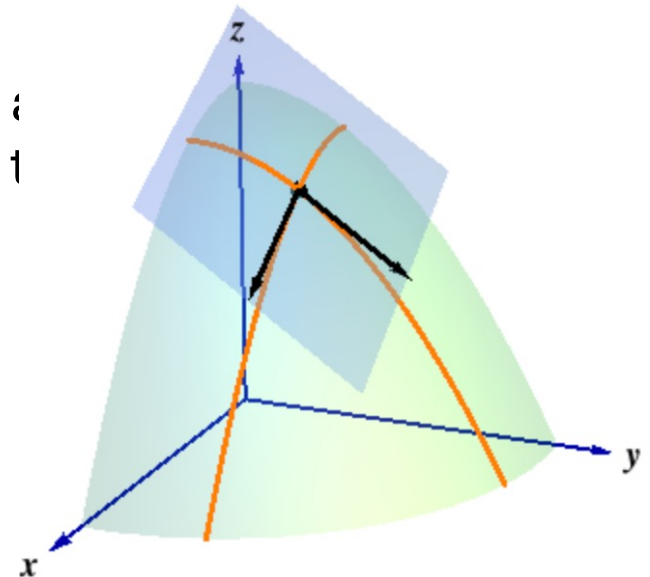
Partial Derivatives



Tangent Plane

Let $P(x_0, y_0)$ be a point on a surface $S, z = f(x, y)$, and let P be any curve passing through P and lying entirely in S . If the tangent lines to all such curves at P lie in the same plane, then this plane is called the **tangent plane** to S at P .

- For a tangent plane to a surface to exist at P , it is sufficient for the function that defines S to be differentiable at that point,



Second-Order Partial Derivatives

When we differentiate a function $f(x, y)$ twice, we produce its second-order derivatives. These derivatives are usually denoted by

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2} & \text{ or } f_{xx}, \text{ defined as } \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) \\ \frac{\partial^2 f}{\partial y^2} & \text{ or } f_{yy}, \text{ defined as } \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) \\ \frac{\partial^2 f}{\partial xy} & \text{ or } f_{xy}, \text{ defined as } \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) \\ \frac{\partial^2 f}{\partial yx} & \text{ or } f_{yx}, \text{ defined as } \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \end{aligned}$$

THEOREM: The Mixed Derivative Theorem

If $f(x, y)$ and its partial derivatives f_x , f_y , f_{xy} , and f_{yx} are defined throughout an open region containing a point (a, b) and are all continuous at (a, b) , then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

The Chain Rule

THEOREM: Chain Rule For Functions of One Independent Variable and Two Intermediate Variables

If $w = f(x, y)$ is differentiable and if $x = x(t), y = y(t)$ are differentiable functions of t , then the composite $w = f(x(t), y(t))$ is a differentiable function of t and

$$\frac{dw}{dt} = f_x(x(t), y(t)) \cdot x'(t) + f_y(x(t), y(t)) \cdot y'(t),$$

or

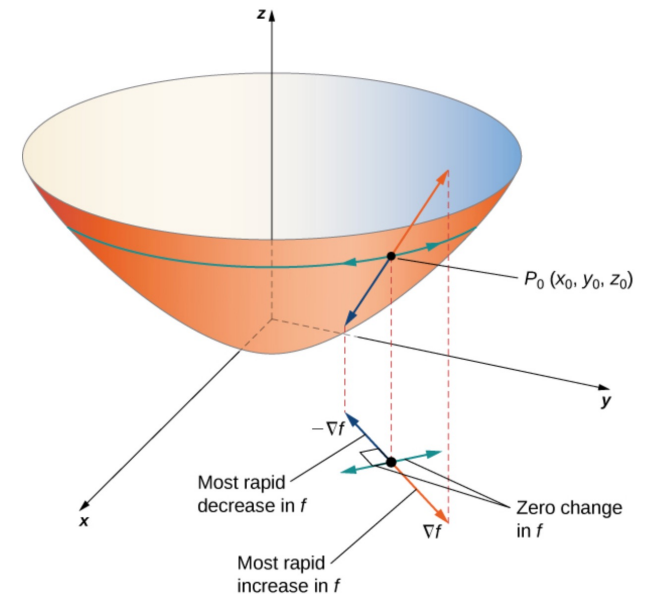
$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Gradient

Let $z = f(x, y)$ be a function of x and y such that f_x and f_y exist. The vector $\nabla f(x, y)$ is called the **gradient** of f and is defined as

$$\nabla f(x, y) = f_x(x, y)\hat{i} + f_y(x, y)\hat{j}.$$

The vector $\nabla f(x, y)$ is also written as “grad f ”.

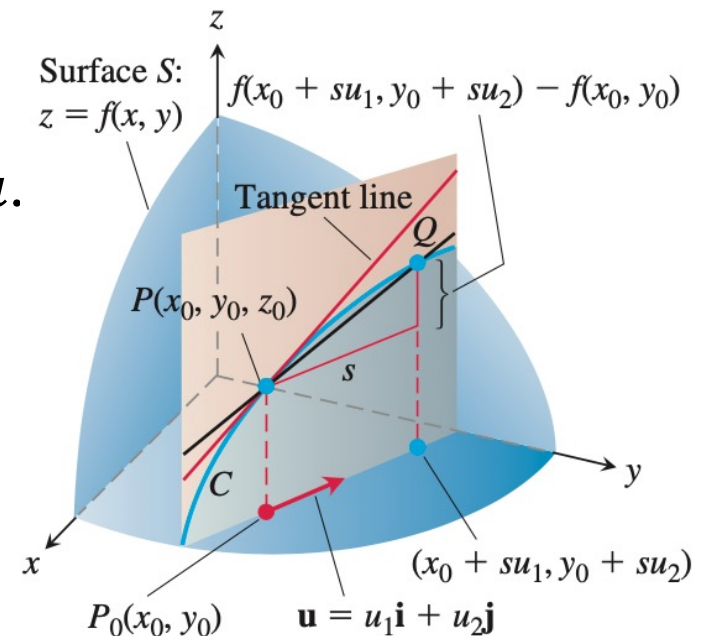


The Directional Derivative

If $f(x, y)$ is differentiable in an open region containing $P_0(x_0, y_0)$, then $(D_{\mathbf{u}}f)_{P_0}$ is the instantaneous rate of change of the surface at P_0 stepping off in the direction $\mathbf{u}$.

$$(D_{\mathbf{u}}f)_{P_0} = (\nabla f)_{P_0} \cdot \mathbf{u}$$

the dot product of the gradient ∇f at P_0 and $\mathbf{u}$.



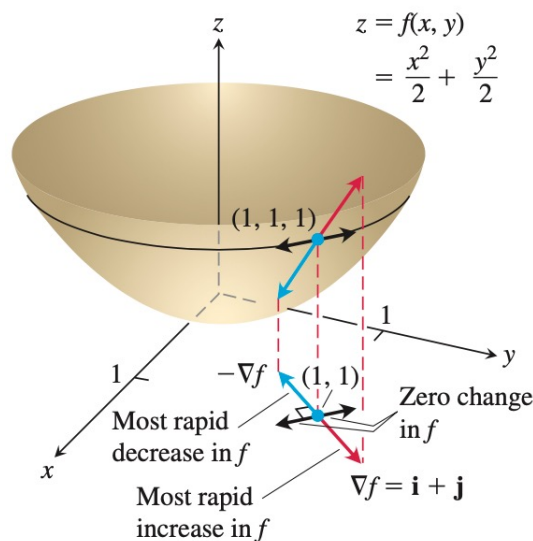


FIGURE 14.30 The direction in which $f(x, y)$ increases most rapidly at $(1, 1)$ is the direction of $\nabla f|_{(1,1)} = \mathbf{i} + \mathbf{j}$. It corresponds to the direction of steepest ascent on the surface at $(1, 1, 1)$ (Example 3).

EXAMPLE 3 Find the directions in which $f(x, y) = (x^2/2) + (y^2/2)$

- (a) increases most rapidly at the point $(1, 1)$, and
- (b) decreases most rapidly at $(1, 1)$.
- (c) What are the directions of zero change in f at $(1, 1)$?

Solution

- (a) The function increases most rapidly in the direction of ∇f at $(1, 1)$. The gradient there is

$$(\nabla f)_{(1,1)} = (x\mathbf{i} + y\mathbf{j})_{(1,1)} = \mathbf{i} + \mathbf{j}.$$

Its direction is

$$\mathbf{u} = \frac{\mathbf{i} + \mathbf{j}}{|\mathbf{i} + \mathbf{j}|} = \frac{\mathbf{i} + \mathbf{j}}{\sqrt{(1)^2 + (1)^2}} = \frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}.$$

- (b) The function decreases most rapidly in the direction of $-\nabla f$ at $(1, 1)$, which is

$$-\mathbf{u} = -\frac{1}{\sqrt{2}}\mathbf{i} - \frac{1}{\sqrt{2}}\mathbf{j}.$$

- (c) The directions of zero change at $(1, 1)$ are the directions orthogonal to ∇f :

$$\mathbf{n} = -\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j} \quad \text{and} \quad -\mathbf{n} = \frac{1}{\sqrt{2}}\mathbf{i} - \frac{1}{\sqrt{2}}\mathbf{j}.$$

See Figure 14.30.

